

Assignment 4

Introduction

For Assignment #4, we will be analyzing the association between the cost of CardiacX, a revolutionary pharmaceutical drug that has the ability to heal cardiac tissue post ST-elevated Myocardial Infarction (STEMI), and various demographic variables. The drug's original indication was for expanding life-span (longevity), but has been repurposed for treatment post-STEMIs. Although results have shown promise in treating patients post-STEMIs, the efficacy of the drug is dose-dependent. Depending on the number of doses required, cost of the drug can vary considerably. Here, we investigate factors that may predict the cost of cardiacX.

Methods

Analysis Plan

We have collected information on 5000 cardiacX users, including their current smoking status (Ever-smoker/Never-smoker), sex, age, whether they used the drug post STEMI, and their cost for CardiacX. For our analysis, we will use R version 4.4.1, using the following packages:

```
#Packages have been loaded in the background
library(tidyverse)
library(table1)
library(data.table)
library(quarto)
library(ggpubr)
library(tinytex)
library(kableExtra)
library(broom)
```

First, our analysis will look at the distribution in cost of cardiacX stratified by Sex, Smoking Status, and STEMI status.

Secondly, we will construct a primary, linear regression model that is the following:

$$Cost = Age + Sex + Smoke + STEMI$$

Our secondary model will add an interaction term between **Smoke** and **STEMI**, to investigate if the cost of CardiacX is exacerbated by the interplay of smoking and STEMI status.

$$Cost = Age + Sex + Smoke + STEMI + Smoke * STEMI$$

Loading in and cleaning dataset

```
raw_data <- fread("cohort.csv")
```

Now we will clean data-set for interpretability moving forward. Females, STEMI, and ever-smoker were initially coded as 1.

```
data_clean <- raw_data |>
  mutate(female_v2 = if_else(female == 0, "Male", "Female"),
         cardiac_v2 = if_else(cardiac == 0, "No", "Yes"),
         smoke_v2 = if_else(smoke == 0, "No", "Yes"))
```

The following is a check for our newly created variables. Output is not included for readability.

```
table(data_clean$female, data_clean$female_v2, useNA = "ifany")
table(data_clean$cardiac, data_clean$cardiac_v2, useNA = "ifany")
table(data_clean$smoke, data_clean$smoke_v2, useNA = "ifany")
```

Labels will be added to our new variables so that figures generated will be easy to interpret

```
label(data_clean$smoke_v2) <- "Smoker"
label(data_clean$cardiac_v2) <- "Cardiac Event"
label(data_clean$female_v2) <- "Sex"
label(data_clean$age) <- "Age"
label(data_clean$cost) <- "Cost"
```

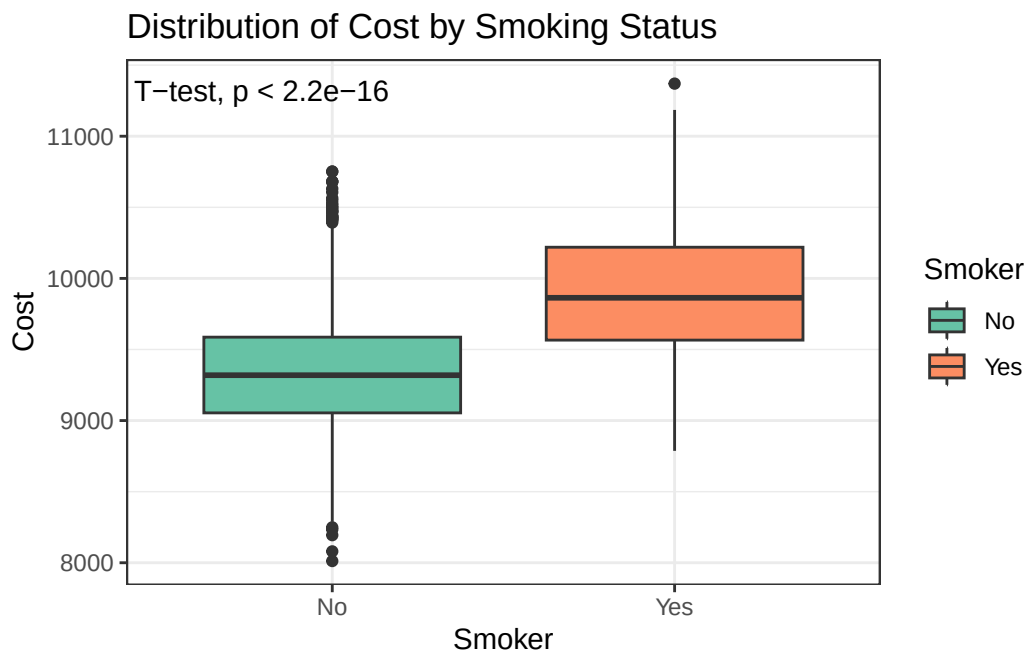
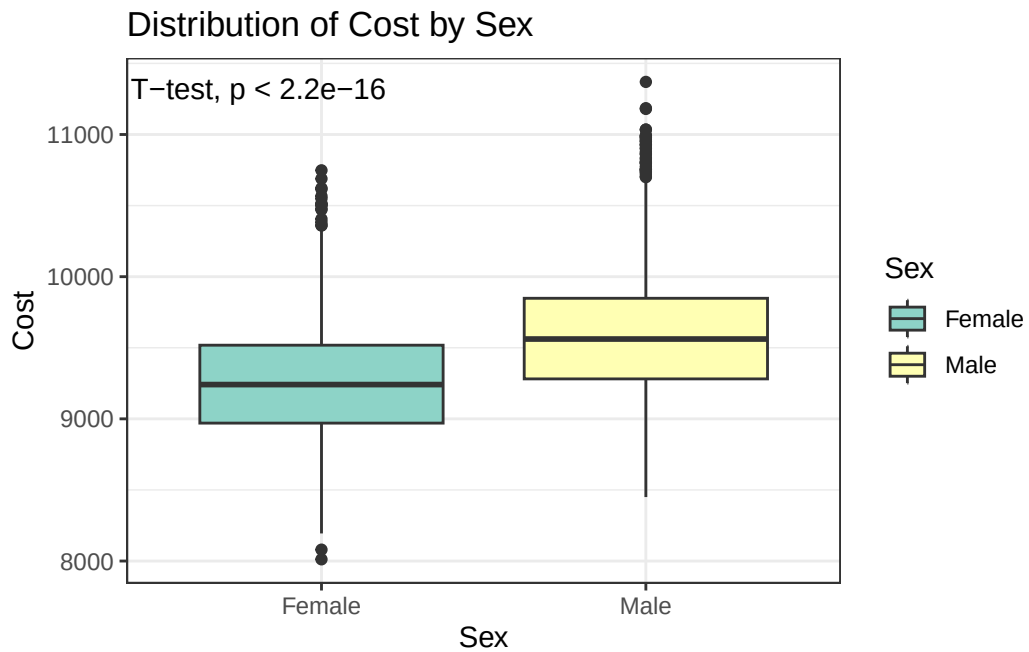
Results

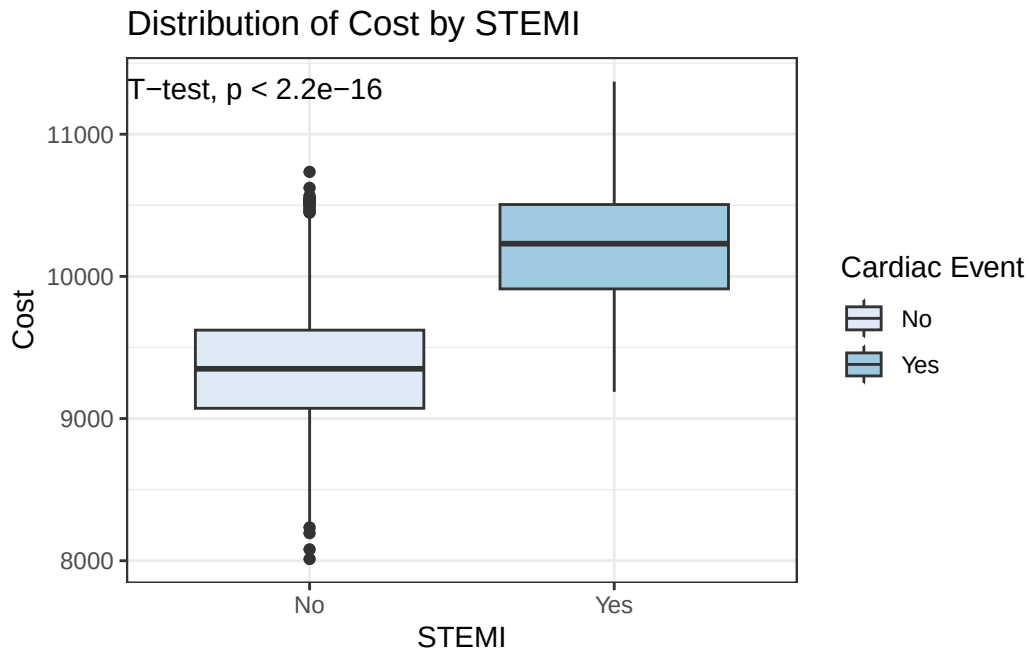
Table of Covariates Stratified by Sex

	Female (N=2866)	Male (N=2134)	Overall (N=5000)
Smoker			
No	2499 (87.2%)	1855 (86.9%)	4354 (87.1%)
Yes	367 (12.8%)	279 (13.1%)	646 (12.9%)
Age			
Mean (SD)	44.7 (15.8)	45.3 (15.5)	44.9 (15.7)
Median [Min, Max]	44.0 [18.0, 72.0]	46.0 [18.0, 72.0]	45.0 [18.0, 72.0]
Cost			
Mean (SD)	9250 (397)	9590 (440)	9400 (448)
Median [Min, Max]	9240 [8010, 10700]	9560 [8450, 11400]	9380 [8010, 11400]
Cardiac Event			
No	2833 (98.8%)	1892 (88.7%)	4725 (94.5%)
Yes	33 (1.2%)	242 (11.3%)	275 (5.5%)

Between males and females, there are similar rates of non-smokers, hovering around 87%. Men have a slightly higher median age of 46 years compared to 44 years for females. Cost was slightly higher for males than females. More men reported using CardiacX for a STEMI (11.3%) compared to women (1.2%).

Differences in distribution of cost for each binary variable





Primary Model

```
model.primary <- glm(cost ~ age + female + smoke + cardiac,
  data = data_clean)
tidy(model.primary)
```

A tibble: 5 x 5

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	8706.	10.7	813.	0
2	age	16.8	0.206	81.4	0
3	female	-271.	6.70	-40.5	5.59e-310
4	smoke	511.	9.97	51.2	0
5	cardiac	520.	15.0	34.6	4.12e-235

Secondary Model

```
model.secondary <- glm(cost ~ age + female + smoke + cardiac + smoke*cardiac,
                      data = data_clean)
tidy(model.secondary)
```

```
# A tibble: 6 x 5
  term          estimate std.error statistic    p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)    8705.      10.7      812.      0
2 age            16.8       0.206     81.4      0
3 female        -271.       6.70    -40.5  6.57e-310
4 smoke          512.      10.7     47.9      0
5 cardiac        524.      19.7     26.5  2.40e-145
6 smoke:cardiac  -8.78      29.5    -0.297  7.66e- 1
```

Discussion

Distribution of cost by demographic variables

We can see that males had a higher average cost of CardiacX compared to females. Those who were smokers had a higher average cost of CardiacX, and those who had a STEMI had a higher cost. Note that these are not adjusted for any potential confounders. Our following linear regression analysis aims to tease out the independent relationship between key, demographic variables.

Linear Regression Models

Our primary revealed that age, sex, smoking status, and STEMI status were all independently associated with the cost of CardiacX. Unsurprisingly, age, smoking status, and STEMI status were associated with a significant increase in the average cost of CardiacX. Being female was associated with a significant decrease in the cost of CardiacX.

Our secondary model revealed that there was no significant interaction between smoking status and STEMI status. This means that being a smoker and using CardiacX for a STEMI does not further increase (or decrease) your cost for CardiacX. The interaction between smoking and STEMI status also did not influence the association between the other covariates and cost of CardiacX.

I did not use generative AI technology (e.g., ChatGPT) to complete any portion of this work.